
On the Geometric Asymmetry of Functional Income Distribution: A Formal Note

0. Motivation and Construction

In the income approach to GDP, the capital and labor components at year i are $C(i)$ and $L(i)$ respectively. Define the auxiliary series $\phi(i) = C(i)/L(i)$ and its inverse $\lambda(i) = L(i)/C(i)$.

Let $\phi^* = \min(\phi(i))$ and $M = \max(\lambda(i)) = 1/\phi^*$ denote the historical extremes: M is the year of maximum relative advantage of Labor, ϕ^* the year of maximum relative advantage of Capital.

Construction of S1A

Fix year i with current capital share $C(i)$. Had the distributional conditions of year M prevailed today, Labor would have received $M \cdot C(i)$ instead of its actual $L(i) = \lambda(i) \cdot C(i)$. The absolute gain of Capital — equivalently, the absolute loss of Labor relative to its best historical conditions — is therefore:

$$\Delta K(i) = (M - \lambda(i)) \cdot C(i)$$

To express this as a *relative* gain, we normalize not by today's economy $C(i) + L(i)$, but by the counterfactual economy as it would have been today under the conditions most favorable to Capital (worst for Labor):

$$GDP^* = C(i) + M \cdot C(i) = (M + 1) \cdot C(i)$$

Dividing $\Delta K(i)$ by GDP^* and simplifying by $C(i)$:

$$S1A(i) = (M - \lambda(i)) / (M + 1)$$

$S1A$ is thus a counterfactual relative measure: the gain of Capital at year i , expressed as a fraction of the economy as it would have been under maximum Labor disadvantage. It is independent of the absolute scale of the economy, since $C(i)$ cancels exactly.

Construction of S1B

The construction follows symmetrically, exchanging the roles of Capital and Labor: the reference year is now ϕ^* (maximum advantage of Capital, minimum of Labor), and the counterfactual denominator is the economy as it would have been under maximum Capital disadvantage. This yields:

$$S1B(i) = (\phi(i) - \phi) / (1 + \phi)^{**}$$

Setting $r = \phi(i)$, both series can be written compactly as:

$$S1A = (r - \phi^*) / r(1 + \phi^*) ; S1B = (r - \phi^*) / (1 + \phi^*) ; S1A = (1/r) \cdot S1B$$

The empirical anchor ϕ^ corresponds to historically meaningful years: USA 1970, UK, Italy and Spain 1976, France 1982 — the peak of the postwar labor-capital compromise in each country, preceding the neoliberal reorientation of the 1980s.*

1. Asymptotic properties

1. $\lim(r \rightarrow 0) S1A = -\infty$
2. $\lim(r \rightarrow \infty) S1A = 1/(1+\phi^*) \leftarrow$ finite upper bound
3. $\lim(r \rightarrow 0) S1B = -\phi^*/(1+\phi^*)$
4. $\lim(r \rightarrow \infty) S1B = +\infty \leftarrow$ unbounded

$S1A$ is bounded above by construction; $S1B$ is unbounded. The biological survival constraint is the only effective limit on $S1B$.

2. The divergence (the wedge)

$$|S1B - S1A| = |(r-1) - (r-\phi^*)| / r(1+\phi^*)$$

The sign depends exclusively on $(r-1)$:

- $r > 1$ (as in Italy) : $S1B - S1A > 0 \rightarrow$ Labor loses more than Capital gains
- $r = 1$: $S1B - S1A = 0 \rightarrow$ both series vanish simultaneously
- $r < 1$: $S1A - S1B > 0 \rightarrow$ Capital loses more than Labor gains

Convexity of the wedge:

$$f(r) = (r-\phi^*)(r-1)/r = r - (1+\phi^*) + \phi^*/r$$

$$f''(r) = 2\phi^*/r^3 > 0 \text{ for all } r > 0, \text{ all } \phi^* > 0$$

The accelerating divergence is a structural property of the functional form, independent of the historical parameter ϕ^* .

3. First derivatives

$$d(S1A)/dr = \phi^* / r^2(1+\phi^*) \rightarrow \text{decreasing in } r, \text{ tends to } 0$$

$$d(S1B)/dr = 1/(1+\phi^*) \rightarrow \text{constant, independent of } r$$

$$\text{The ratio: } [d(S1A)/dr] / [d(S1B)/dr] = \phi^*/r^2 \rightarrow 0 \text{ as } r \rightarrow \infty$$

4. Second derivatives

$$\partial^2 S1A / \partial r^2 = -2\phi^* / r^3(1+\phi^*) < 0 \rightarrow \text{strictly concave}$$

$$\partial^2 S1B / \partial r^2 = 0 \rightarrow \text{linear}$$

5. Three structural correspondences with Marx

5.1 — The General Law of Capitalist Accumulation (polarization)

Marx (*Capital*, Vol. I, Ch. 25): "Accumulation of wealth at one pole is, therefore, at the same time accumulation of misery, agony of toil, slavery, ignorance, brutality, mental degradation, at the opposite pole."

The wedge $(r-1)(r-\phi^*)/r(1+\phi^*)$ is a theorem, not a metaphor. For $r > 1$, polarization is convex and accelerating in r , for any value of the historical parameter ϕ^* .

5.2 — The Tendency of the Rate of Profit to Fall (satiation of Capital)

Marx (*Capital*, Vol. III, Ch. 13): the progressive difficulty of extracting proportional surplus value as the system expands.

$\partial^2 S1A/\partial r^2 = -2\phi^*/r^3(1+\phi^*) < 0$: the marginal relative gain of Capital is strictly decreasing for all r and all ϕ^* . The deceleration is structural.

5.3 — The Biological Exhaustion of Labor Power (the "Vampire")

Marx (*Capital*, Vol. I, Ch. 10): Capital "takes no account of the health and the length of life of the worker" and acts as a vampire with no intrinsic limit, stopping only when the host organism is exhausted.

$d(S1B)/dr = 1/(1+\phi^*)$: the marginal extraction rate is constant — it does not decelerate with increasing r . The only effective brake is the biological constraint (demographic collapse).

6. Summary

The geometry of $S1A$ and $S1B$ encodes a universal asymmetry: the party in distributional disadvantage faces potentially unlimited relative losses, while the party in advantage faces a structurally bounded relative gain. This wedge not only exists but accelerates convexly with distributional imbalance — for any ϕ^* , for any country. The three core tendencies identified by Marx in *Capital* emerge as analytical properties of normalized national accounts series.